XINRAN ZHANG

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PROFILE

My research focuses on human–environment relationships through GIScience, spatial analysis, and computational methods. I am particularly interested in developing interpretable and reusable computational methods for understanding spatial and temporal relationships, bridging methodological innovation with practical research applications.

EDUCATION

University of Edinburgh, Edinburgh College of Art **Sep 2023-Sep 2024**

MSc in Landscape and Wellbeing (Merit) *Edinburgh, UK*

Dissertation: Soundscape Perception and Social Interaction in Urban Blue Spaces: A Mixed-Methods Study (82/100, Distinction)

Fuzhou University, School of Architecture and Urban Planning **Sep 2017-Jun 2022**

BEng in Landscape Architecture *Fuzhou, CN*

Undergraduate Thesis: GIS-Based Noise Mapping and Soundscape Informed Urban Park Design (Top 2 in cohort)

RESEARCH EXPERIENCE

AFTERMATH: Evaluating Terminal-State Externalities in Embodied Spatial Planning **Jul 2026–Present**

Independent Researcher & Developer | GitHub | Overview

- Developed a reproducible benchmark examining how robot actions reshape accessibility for subsequent users across 20 indoor scenes, three mobility profiles, and nine route demands per terminal state.
- Implemented a deterministic manipulation simulator and exact A* search for robot-cost optimality auditing; evaluated ~1,220 planning episodes and 90 unique terminal-state configurations.
- Found accessibility harm in 18/20 robot-efficient baseline states and identified improvements in eight scenes within a 15% robot-cost allowance, revealing trade-offs between task efficiency and downstream accessibility.

HumanRoute: Evaluating Pedestrian Routes with Vision–Language Models **May 2026-Present**

Independent Research | Overview

- Developed a VLM evaluation pipeline using Mapillary imagery from 13 Seattle pedestrian routes (~8 km) to investigate context-dependent environmental assessments.
- Designed multi-scale sampling (12/24/48 m) and contextual inputs (3/5/9 frames) to examine how spatial information influences model judgments.
- Implemented six-dimensional assessments conditioned on adult, older-adult, and wheelchair/stroller user profiles to compare accessibility-related judgments.

Blue-Space Soundscape Restoration and Social Behaviour: A Mixed-Methods Study **Mar 2024-Aug 2024**

Master's dissertation, supervised by Prof. Simon Bell, ***(Distinction, 82/100)*** *|* Overview *Glasgow, UK*

- Investigated relationships between soundscape perception, restorative experiences, and social behavior in urban blue spaces through mixed-methods field research.
- Conducted soundscape assessments at 65 locations, recorded 1,780 behavioral observations, and interviewed 55 participants.
- Integrated spatial exposure mapping, regression analysis, and qualitative interpretation to examine environmental influences on staying and social interaction, including age-related differences in perception.

- **StreetSignals: Computational Representation of Human-Centered Urban Environments** **Apr 2025-Sep 2025**

Independent Methodological Pilot Study, inspired by Exposome-NL programme | Overview *Remote*

- ✦ Developed a GeoAI framework combining ResNet50-based street-view semantic segmentation with interpretable indicators of visual composition, traffic dominance, and spatial enclosure.
- ✦ Designed structural indicators of connectivity, clustering, and spatial organization to explore scalable representations of pedestrian-relevant environmental conditions and potential movement constraints.

RESEARCH SOFTWARE & TOOL DEVELOPMENT

Grain Checker: QGIS Plugin for MAUP Sensitivity Diagnostics **Feb 2026-Mar 2026**

Research Tool Development (Independent) |Github| Demo video

- ✦ Developed a QGIS plugin to assess the sensitivity of bivariate spatial relationships to aggregation scale, supporting reproducible diagnostics of the modifiable areal unit problem (MAUP).

Parcel-Level Land-Use Continuity: A QGIS Tool for Life-Course Exposure Reconstruction **Apr 2025-Jul 2025**

Research Tool Development (Independent) |Github | Demo video:

- ✦ Developed a QGIS tool to reconstruct historical parcel-level land-use continuity across changing spatial boundaries, supporting longitudinal environmental exposure analysis.

TECHNICAL & ANALYTICAL SKILLS

GIScience & Spatiotemporal Analysis: Spatial statistics, spatial interpolation (kriging), spatial accessibility analysis, longitudinal land-use reconstruction, historical parcel matching, spatial aggregation and MAUP sensitivity analysis, space syntax, geospatial data processing.

Geospatial Computing & Research Software: Python (NumPy, geospatial data processing, algorithm implementation), QGIS plugin development, GIS workflow automation, reproducible computational experiments, graph-based pathfinding (A*), computational geometry.

GeoAI & Spatial Data Representation: Street-view semantic segmentation (ResNet50), computer vision, vision-language model evaluation, multimodal prompting, spatial structural indicators, multi-scale sampling, geospatial visualization.

Empirical Research & Software Tools: Behavioral observation, semi-structured interviews, mixed-methods analysis; QGIS, ArcGIS, R, Stata, NVivo, AnyLogic, Cadna/A, PyTorch, Git/GitHub.

AWARDS & CONFERENCE PRESENTATIONS

National Award & Oral Presentation, 2nd National Soundscape Academic Forum of China **Oct 2022**

Best Undergraduate Project Award (Top 6%), School of Architecture and Urban Planning of FZU **Jun 2022**

National Prize (Top 2.6%, 18/784 entries), National Urban-Rural Planning Competition for Undergraduates **Dec 2020**

PROFESSIONAL EXPERIENCE

Research Assistant, Fuzhou University & Collaborative Labs **Sep 2019-Mar 2024**

- ✦ Supported faculty-led projects on soundscape perception and urban informality through GIS analysis, fieldwork, and research reporting. Delivered GIS and remote sensing training workshops for undergraduate design competitions.

SOCIAL ENGAGEMENT

Independent knitwear designer, Ravelry **2026**

Student Advisor, Design Competitions, School of Architecture and Urban Planning, FZU **Fuzhou, CN, 2021**

Volunteer, Welfare Home for Individuals with Cognitive Impairments **Fuzhou, CN, 2017-2019**

- ✦ Supported daily services and developed early awareness of sensory processing differences and accessibility needs.